7. **(a)** In a nuclear reaction, a uranium-235 $\left({}^{235}_{92}\text{U}\right)$ nuclide is transformed into an unstable uranium-236 nuclide $\left({}^{236}_{92}\text{U}\right)$ through bombardment by a slow-moving neutron. The unstable uranium-236 nuclide undergoes nuclear fission to form stable products of a lathium-139 nuclide $\left({}^{139}_{57}\text{La}\right)$ and a nuclide of bromine $\left({}^{A}_{Z}\text{Br}\right)$.

$$ {}^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{236}_{92}\text{U} \rightarrow {}^{139}_{57}\text{La} + {}^{A}_{Z}\text{Br} + 3\,{}^{1}_{0}\text{n} + \gamma $$

Use the following masses in answering this question:

rest mass of ${}^{235}_{92}\text{U}$ nuclide	= 235.044*u*
rest mass of ${}^{139}_{57}\text{La}$ nuclide	= 138.906*u*
rest mass of proton	= 1.00728*u*
rest mass of neutron	= 1.00866*u*

(i) Determine the nucleon number *A* and proton number *Z* of the bromine nuclide.

A = ...

Z = ... [1]

(ii) Calculate the binding energy of the uranium-235 nuclide to 4 significant figures.

binding energy = ... MeV [3]

(iii) State the feature of this equation that indicates that a chain reaction may be possible.

.....

[1]

(b) In a laboratory source of strontium-90, the number of atoms present in the year 2013 was 2.36×10^{13}. Strontium-90 decays by emission of a β-particle and this nuclide has a half-life of 28 years.

(i) State what is a β-particle.

.....

[1]

(ii) Calculate the activity of the source in the year 2113.

activity = ... Bq [2]